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Subject Code: DM-303
Paper ID: 13003

GNIOT Institute of Management Studies – PGDM (Batch 2022-24)

(TRIMESTER – III) END TERM EXAMINATION, APRIL - 2023

ANALYTICS FOR MANAGERS

[Time: 2 Hours 30 Minutes] ***[Maximum Marks: 100]***

INSTRUCTIONS:
(i) ***There are THREE sections in this question paper.***
(ii) ***Attempt questions from each section as per the directions.***
(iii) ***Make suitable assumptions wherever necessary.***

SECTION-A

(Short Answer Type Questions, do not exceed 100 words)

Q.1. Write short notes on the following: ***(10×3=30)***

(a) What is the difference between a list and a tuple in Python?

(b) What is the difference between "while" and "for" loops in Python?

(c) Training Vs Testing data

(d) Explain Balanced Vs Imbalanced Data.

(e) Write steps to install pandas.

(f) Explain Supervised and Unsupervised learning.

(g) What is the use of numpy

(h) Explain Array with syntax.

(i) How do you use comment in a Python program, write syntax for single and multiple line comment.

(j) What is Business Analytics?

SECTION-B

(Long Answer Type Questions)

Attempt all questions of the following: ***(4×10=40)***

Q.2. *(a)* Write a program to find largest number in a list.

OR

*(b)*What are the techniques of supervised and unsupervised learning?

Q.3. *(a)* What are the major steps of data Pre-Processing?

OR

(b) Write a program to print duplicates from a list of integers.

Q.4. *(a)* What is predictive modeling*?*

OR

(b) Write a program to check whether the number is prime or not?

Q.5. *(a)* List the phases of CRISP DM process for Data Analysis.

OR

(b) Explain the concept of Regression vs. classification.

SECTION-C

Q.6. Attempt all parts of the following: ***(30)***

Let us say you are working for a company that sells clothing online, and you want to analyze customer behavior to improve sales. You have a dataset CUSTOMER_PURCHASES containing information about customer purchases, including the product name, price, quantity and date of purchase. Your goal is to use Python to analyze the data and answer these questions:

(5×3=15)

6(a)

i. Display Top 10 Rows of The Dataset
ii. Display Last 10 Rows of The Dataset
iii. Find Shape of Our Dataset (Number of Rows and Number of Columns)
iv. Get Overall Statistics About the Data Frame
v. Product name having Maximum price

6(b) ***(5×3=15)***

i. What are the top 10 most popular products?
ii. Adding new Column MOBILE_NO to existing data frame by using data frame. insert ()
iii. What is the average price of products?
iv. List of the unique product
v. Sum of null value in the Dataset.